

Design and Synthesis of Half Adder using Yosys

INTRODUCTION:

In this lab, students will learn to synthesize a Half Adder circuit using Yosys. The design takes two inputs (A and B) and produces Sum and Carry outputs. Students will understand how the Sum is generated using an XOR gate and the Carry using an AND gate, making the Half Adder a fundamental building block for arithmetic circuits such as Full Adders and binary adders.

APPLICATION:

Half adders are used for basic binary addition in ALUs, digital calculators, and processors. They also serve as building blocks for full adders, binary counters, and other arithmetic circuits due to their simplicity and efficiency.

PROCEDURE TO CREATE A LAB:

Step 1: Set up Project Directory

- 1.1: Create a directory for your lab: `mkdir <project_name>`
- 1.2: Open lab directory: `cd <Project_name>`
- 1.3: Create directory for project: `mkdir halfadder`
- 1.4: Open project directory: `cd halfadder`

Step 2: Create Design file

Create Design file: `gvim half_adder.v`

RTL Design Code:

```
module half_adder(  
    input a, b,  
    output sum, cout );  
    assign sum = a ^ b;  
    assign cout = a & b;  
endmodule
```

Step 3: Synthesize the Design with Yosys

invoke Yosys in a new terminal by the following command
yosys

3.1: Use the following command to

Synthesize:

```
# Read the Verilog source file
read_verilog half_adder.v

# Set the top-level module and check hierarchy
hierarchy -check -top half_adder

# high level stuff
proc; opt; memory; opt;

# mapping to internal cell library
techmap; opt

# mapping logic to mycells.lib
abc -liberty osu018_stdcells.lib

# Clean up unused cells and signals after mapping
clean

# Write the final synthesized gate-level netlist to a new Verilog file
write_verilog half_adder_synth.v

#Generate a graphical visualization of the synthesized circuit
show
```

Step 4: View Output File (Netlist)

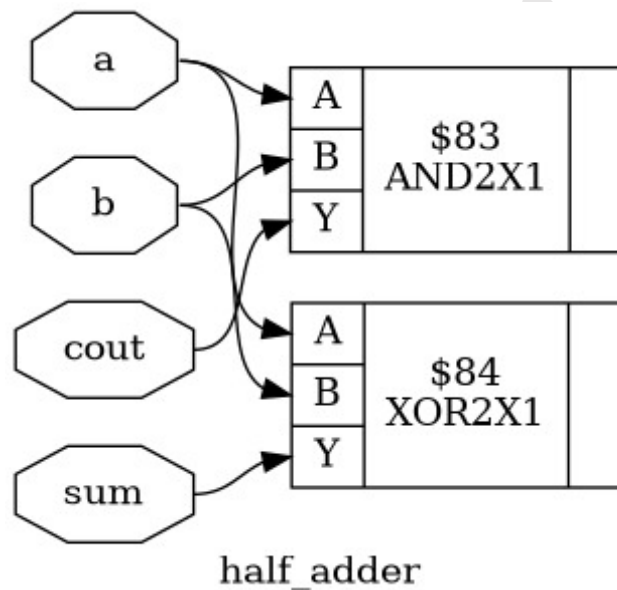
```
gvim half_adder_synth. v
```

OBSERVATION:

- The Half Adder design is synthesized without any errors.
- Basic logic gates (XOR and AND) are correctly utilized to generate the Sum and Carry outputs.
- The Sum output is accurately derived using XOR logic between the two input bits.

- The Carry output is generated using AND logic, representing the carry-over in binary addition.
- The circuit performs binary addition of two single-bit inputs effectively and can be used as a building block for more complex arithmetic circuits like the Full Adder.

OUTPUT:



CONCLUSION:

The Yosys synthesis output confirms the correct implementation of the Half Adder circuit. The RTL design is successfully read, optimized, and mapped to the target standard cell library. The generated gate-level netlist preserves the intended logic, with XOR and AND gates accurately realizing the Sum and Carry outputs, respectively.